Jaume Puig

Space Radiobiologist and Engineer

jaume.puig98@gmail.com | jaumepuig98.github.io | linkedin.com/in/jaume-puig
ORCID 0000-0002-3642-4919

PROFESSIONAL EXPERIENCE

Oct 2024 to Present
CTO & Co-Founder
Barding Industries
- Architecting the technology platform: engineered biology that recovers critical minerals, including rare earth elements, from ores and waste streams.

Jun 2024 to Present
Communications Coordinator & Member, Microbes & Hardware AWGs
NASA Open Science Data Repository
- Contributing to open-science space microbiology through omics data reanalysis, pipeline optimization, and methods standardization.
- Co-authoring the SPLASH Database paper alongside AWG principal investigators.
- Managing internal communications and inter-subgroup coordination across the AWG.

Feb 2021 to Present
Judge, Mentor & Community Member
iGEM Foundation
- Executive Judging Committee Member (2026 to Present): guiding the evolution and quality of judging across the competition, from judging philosophy, standards, and rubrics to training incoming judges and ensuring fairness and consistency.
- Space Network Member (2024 to Present): advising teams on space exploration and astrobiology applications.
- Judge (2024 to Present): evaluating international synthetic biology projects.
- Mentor (2022 to Present): technical guidance across experimental design and modeling.
- Ambassador for Europe (2022 to 2024): promoted synthetic biology across Europe.

May 2024 to Present
STEAM Educator
AulaSPUTNIK
- Designing and delivering science workshops for children, and managing the organization's website.

Jun 2024 to Jan 2025
Research Fellow, Crew 2: The Illuminator's Class
BioAstra
- Analyzed pharmacogenomic and nutritional genomics datasets from spaceflight missions to advance commercial astronaut health research.
- Contributed to protocol development for the Space Omics and Medical Atlas (SOMA) framework.

Apr 2023 to Jan 2024
Space Biotechnology Technician
Yuri GmbH
- Conducted microbial culture experiments and multi-omics analysis for ISS payload projects using automated modular bioreactors.
- Managed laboratory facilities, hardware testing, and data interpretation across multiple concurrent payload programs.

ACADEMIC BACKGROUND

2025 to Present
Ph.D. in Biomolecular Sciences
Laurentian University / SNOLAB
- Major: Radiation Biology.
- McLaughlin Graduate Scholarship recipient.
- Teaching Assistant.

2022 to 2023 **M.Sc. in Multidisciplinary Research in Exp. Sciences**
BIST / UPF
- Major: Low-Background Radiation Omics. Minor: Ancestral Gene Resurrection.
- JAE Intro ICU Grant (CSIC).
- Lead organizer of ScienceHack 2023 (BIST-MMRES), a science hackathon with 360+ participants.

2016 to 2022 **B.Eng. in Biomedical Engineering**
Universitat Pompeu Fabra (UPF)
- Major: LEO Radiation Biotolerance.
- iGEM 2020: Gold Medal, Best New Basic Part, Best Wiki. Nominated for Best Therapeutics Project, Best Poster, and Best Presentation.

PUBLICATIONS

2025 **STREAMS guidelines: standards for technical reporting in environmental and host-associated microbiome studies**
Kelliher, Mirzayi, Bordenstein, … STREAMS Consortium, et al. Nature Microbiology 10, 3059–3068.

2025 **Feeding the cosmos: tackling personalized space nutrition and the leaky gut challenge**
Barbero Barcenilla, Rivero, Lynch, … **Puig**, et al. npj Microgravity 11, 45.

2024 **Intein-mediated thyroid hormone biosensors: towards controlled delivery of hormone therapy**
Martí-Baena, Pascuet-Fontanet, Berjaga-Buisan, Caravaca-Rodríguez, **Puig-Costa-Jussà**, et al. Frontiers in Systems Biology 4:1270071.

2021 **DNA damage protection for enhanced bacterial survival under simulated low Earth orbit environmental conditions in Escherichia coli**
Puig, Knödlseder, Quera, Algara, Güell. Frontiers in Microbiology 12:789668.

2024 **Space Omics and Medical Atlas (SOMA) protocol compilation**
Contributor. Open protocols for spaceflight omics and medical research.

2020 **Hormonic: intein-mediated thyroid hormone biosensors**
iGEM 2020, Team UPF_Barcelona wiki. Gold Medal, Best New Basic Part, Best Wiki.

PRESENTATIONS

Oct 2026 **Global consistency of biological responses to sub-terrestrial low-background radiation environments**
Radiation Research Society (RRS) 72nd Annual Meeting, Waikoloa, Hawaii, USA.

Oct 2024 **Spaceflight effects on human skin microbiome**
International Astronautical Congress (IAC) 2024, Milan, Italy.

Oct 2023 **DNA repair and genetic radiation biotolerance**
International Astronautical Congress (IAC) 2023, Baku, Azerbaijan.

Sep 2022 **DNA damage protection for enhanced bacterial survival under simulated low Earth orbit conditions**
Europlanet Science Congress 2022, Granada, Spain.

May 2022 **DNA damage protection for enhanced bacterial survival under simulated low Earth orbit conditions**
AGU / NASA Astrobiology Science Conference (AbSciCon), Atlanta, USA.

AWARDS & HONORS

2025 **McLaughlin Graduate Scholarship** Laurentian University.

2022 to 2023 **JAE Intro ICU Grant** Spanish National Research Council (CSIC), awarded during the M.Sc.

2022 **NASA Space Apps Challenge Finalist** Presented at the NASA TOPS Forum (virtual, April 2023): “Measuring Open Science”.

2020 **iGEM Gold Medal, Best New Basic Part, Best Wiki** Team UPF_Barcelona. Also nominated for Best Therapeutics Project, Best Poster, and Best Presentation.